Hands-On Lab: Comparing AI Tools

"Same task. Five tools. Who does it best?"

1. Header Information

- **Title:** Hands-On Lab: Comparing AI Tools ("Same task. Five tools. Who does it best?")
- **Student Name:** Adarsh Bairagi
- **Course & Year:** IMCA 2nd Year
- **Date:** October 13, 2026
- **AI Tools Selected:** ChatGPT, Gemini, Claude

2. Task Execution & Responses

Task 1: Explain Task

- **Prompt:** *Explain how UPI payments work to a 15-year-old, in 150 words.*

Tool 1 Output (ChatGPT)

Imagine UPI (Unified Payments Interface) as a digital bridge between your bank account and anyone else's, instantly moving money using just a smartphone. Instead of carrying cash or typing long account numbers and IFSC codes, UPI lets you link your bank account to a unique Virtual Payment Address (VPA) or simple mobile number.

When you buy a snack or send money to a friend, you scan their QR code or enter their ID, type in the amount, and enter your secret 4 or 6-digit UPI PIN. The system immediately verifies your PIN, transfers the funds directly from your bank to theirs within seconds, and sends both of you a confirmation message. It works 24/7, making digital payments as fast and easy as sending a chat message!

Tool 2 Output (Gemini)

UPI stands for Unified Payments Interface, and it's basically an instant, real-time payment system that connects all your bank accounts into a single mobile app. Think of it like sending a text message, but instead of words, you are sending money!

To pay someone, you simply open a UPI app (like Google Pay or PhonePe), scan a vendor's QR code or type in a unique UPI ID, and hit send. Before the transfer completes, you type in a secure UPI PIN—a secret code only you know. Once verified, money moves directly from your bank account to theirs in just seconds, day or night. You don't need to enter long bank account details or wait for approvals!

Tool 3 Output (Claude)

Think of UPI as an instant messaging app, but for money instead of texts. Normally, transferring bank money requires complex account numbers and codes, but UPI simplifies this by connecting your bank directly to a simple app on your phone.

When you want to pay for something, you scan a QR code or enter a unique UPI ID (like an email address for your money). You enter the amount and type in your secret UPI PIN to approve it. Seconds later, the funds securely hop from your bank account right into the receiver's account. It works 24/7 without extra fees, making paying for things fast and effortless.

Task 2: Research Task

- **Prompt:** *What are three current career paths in my field that use AI? Give sources.*
  (Tailored for Integrated Master of Computer Applications - IMCA)

Tool 1 Output (ChatGPT)

1. **AI / Machine Learning Engineer:** Focuses on designing, building, and deploying machine learning models into production software systems and enterprise applications.
    - *Source:* Gartner Research – *Top Strategic Technology Trends*.
2. **Data Scientist & AI Analytics Consultant:** Uses AI models and statistical algorithms to analyze large datasets, optimize application performance, and provide data-driven insights.
    - *Source:* U.S. Bureau of Labor Statistics (BLS) – *Occupational Outlook Handbook: Data Scientists*.

3. **AI Solutions Architect:** Designs scalable cloud architectures and software systems that integrate Large Language Models (LLMs) and computer vision APIs into enterprise software.
    - *Source:* McKinsey & Company – *The State of AI*.

Tool 2 Output (Gemini)

1. **Machine Learning Operations (MLOps) Engineer:** Specializes in bridging software engineering with AI by automating model deployment, continuous monitoring, and infrastructure scalability.
    - *Source:* Google Cloud Architecture Center – *ML Ops Framework Guide*.
2. **Full-Stack AI Developer:** Builds end-to-end web and mobile applications powered by AI services, APIs, and custom model backends.
    - *Source:* GitHub Tech Trends – *The Future of AI-Powered Software Engineering*.
3. **AI Ethics & Security Specialist:** Focuses on auditing AI algorithms for security vulnerabilities, compliance with privacy regulations, and fairness in automated decision systems.
    - *Source:* World Economic Forum (WEF) – *Global Technology Governance Report*.

Tool 3 Output (Claude)

1. **AI Research Scientist:** Conducts core research into advancing neural network architectures, NLP models, and computer vision algorithms.
    - *Source:* Association for Computing Machinery (ACM) – *Digital Library*.
2. **NLP (Natural Language Processing) Engineer:** Specializes in creating algorithms and pipelines that process, interpret, and generate human languages within application software.
    - *Source:* Stanford Institute for Human-Centered AI (HAI) – *AI Index Report*.
3. **Computer Vision Engineer:** Develops intelligent systems for automated image analysis, video processing, and augmented reality applications.
    - *Source:* MIT Technology Review – *Insights on Machine Perception*.

Task 3: Writing Task

- **Prompt:** *Write a polite email to my professor asking for a deadline extension of two days.*

Tool 1 Output (ChatGPT)

Subject: Extension Request: [Course Name] Assignment Submission – Adarsh Bairagi

Dear Professor [Last Name],

I hope this email finds you well.

I am writing to respectfully request a brief two-day extension for the upcoming [Assignment Name], which is currently due on [Original Due Date]. Due to [briefly state reason, e.g., an unexpected personal illness / overlapping academic project deadlines], I am finding it challenging to complete the work to the standard I aim for.

With an extension until [New Proposed Date], I will be able to submit a thorough and polished assignment. I have already completed [mention completed part, e.g., the initial research] and am working diligently on the remaining sections.

Thank you very much for considering my request. I appreciate your time and understanding.

Best regards,
Adarsh Bairagi
IMCA 2nd Year

Tool 2 Output (Gemini)

Subject: Adarsh Bairagi (IMCA 2nd Year) – Request for two-day extension on [Assignment Name]

Dear Professor [Last Name],

I hope you are having a good week.

I am reaching out to inquire if it might be possible to get a two-day extension on the [Assignment Name] due on [Original Due Date]. Unfortunately, [insert reason, e.g., a sudden illness / unforeseen circumstances] has set me back on my schedule over the past few days.

I am eager to submit my best quality work for your class, and having until [New Proposed Date] would allow me the time needed to properly finalize my draft.

Please let me know if this is acceptable. I appreciate your guidance and consideration!

Warm regards,
Adarsh Bairagi
IMCA 2nd Year

Tool 3 Output (Claude)

Subject: Request for Deadline Extension – [Assignment Title] – Adarsh Bairagi

Dear Professor [Last Name],

I hope you are doing well.

I am writing to ask if I might be granted a 2-day deadline extension for [Assignment Title], currently due on [Due Date]. Over the past few days, I have encountered [unexpected illness / personal conflict], which has limited my ability to complete the assignment to the best of my ability.

If approved, I would submit the completed work by [New Date & Time]. I am fully committed to completing this assignment thoroughly and would be very grateful for the extra time to ensure its quality.

Thank you for your time, understanding, and consideration.

Sincerely,
Adarsh Bairagi
IMCA 2nd Year

3. Comparison & Scoring Table

Score each tool from 1 (Poor) to 5 (Excellent) based on testing:

Evaluation Metric	ChatGPT	Gemini	Claude
Accuracy	5/5	5/5	5/5
Sources	4/5	4/5	4/5

Evaluation Metric	ChatGPT	Gemini	Claude
Clarity	5/5	5/5	5/5
Speed	5/5	5/5	4/5
Total Score	**19/20**	**19/20**	**18/20**

4. Reflection & Key Takeaways

- **Surprise / Key Finding:** While all three AI tools provided clear, high-quality responses for each prompt, ChatGPT and Gemini stood out for their speed and responsiveness. For the research task tailored to IMCA/Computer Applications, Claude cited highly academic research platforms like ACM and Stanford HAI, whereas ChatGPT and Gemini provided practical, industry-oriented roles and sources like Google Cloud and Gartner. Overall, all three tools effectively demonstrated strong performance across explanation, research, and functional writing tasks.